

EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND INFORMATION TECHNOLOGY



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FEASIBILITY STUDY

EQUIPMENT BORROWING MANAGEMENT SYSTEM (EBMS)

EASTERN INTERNATIONAL UNIVERSITY

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1. EXECUTIVE SUMMARY

Eastern International University (EIU) currently manages the process of borrowing and returning teaching equipment — such as projectors, laptops, microphones, and tablets — manually using paper-based forms. This manual process often leads to data inaccuracy, delays, loss of records, and difficulty in tracking overdue items.

To resolve these issues, the IT Department proposes developing a web-based Equipment Borrowing Management System (EBMS) to automate the entire process. The new system will enable faculty and students to submit borrowing requests online, receive digital approval, and check real-time equipment availability.

The total estimated project cost is VND 120,000,000, with a projected implementation time of five months. Financial analysis indicates a return on investment (ROI) of approximately 96.6% and a payback period of about 1.2 years. The project is technically feasible, economically justified, and aligned with EIU's digital transformation initiative. Therefore, it is strongly recommended for implementation.

2. DESCRIPTION OF PRODUCTS AND SERVICES

The Equipment Borrowing Management System (EBMS) is a web-based application that automates the borrowing and returning process for teaching equipment. It will replace the outdated manual method with an integrated digital solution accessible via web browsers on computers and mobile devices.

Main Features

- User Authentication and Management: Role-based access for administrators, IT staff, and general users (faculty, students).
- Online Request Submission: Users can browse available equipment and send requests digitally.
- Approval Workflow: IT staff or department heads can approve, reject, or reschedule borrowing requests online.
- Real-Time Tracking: Monitors the status of all equipment — available, borrowed, reserved, or under maintenance.
- Notification System: Sends email or in-app reminders for upcoming due dates and overdue items.
- Reports and Analytics: Generates monthly and annual statistics on borrowing frequency, popular items, and overdue records.
- Activity Log (Audit Trail): Records all user activities for security and traceability.

Benefits

- Reduces time spent on manual paperwork.

- Improves accuracy and accountability.
- Provides transparent, up-to-date data for IT staff and administrators.
- Contributes to EIU's sustainability goals by reducing paper use.

3. TECHNOLOGY CONSIDERATIONS

The EBMS will be built using modern, open-source, and scalable technologies that align with EIU's IT infrastructure.

Component	Technology	Description
Frontend	HTML5, CSS3, JavaScript	Responsive interface compatible with browsers and mobile devices.
Backend	Python (FastAPI Framework)	Efficient, asynchronous server-side processing and API handling.
Database	PostgreSQL	Open-source relational database for structured data management.
Hosting	EIU's internal server or cloud (Render/Azure)	Ensures scalability and availability.
Security	HTTPS, JWT authentication, password hashing	Protects user data and maintains secure access control.

EIU's IT department possesses sufficient expertise to deploy and maintain this system with minor external consultation. The expected additional cost for server hosting and maintenance is approximately 8–10% of the total annual IT budget.

Overall, the technical risk level is low, as all chosen technologies are mature, well-documented, and widely used.

4. PRODUCT/SERVICE MARKETPLACE

The The EBMS is not a commercial product but an internal operational tool designed to serve EIU's faculty, staff, and students. However, it reflects the broader trend of digital transformation in education management systems.

Across global and regional universities, many institutions are adopting similar solutions to:

- Improve efficiency in managing teaching resources.
- Minimize equipment loss and misuse.
- Increase accountability in asset tracking.

At EIU, EBMS will be the pilot digital management system that can later expand to cover other university assets such as laboratory equipment, office supplies, and library materials.

By leading this digital initiative, EIU enhances its reputation for innovation and sustainability within Vietnam's higher education ecosystem.

5. MARKETING STRATEGY

As an internal project, EBMS does not require external marketing. Instead, an internal communication and adoption strategy will be implemented to ensure user engagement and system acceptance.

Promotion and Training Plan

1. Awareness Campaign: Announce the project through university email, social media pages, and notice boards.
2. Demonstration Workshops: Conduct brief demo sessions during departmental meetings.
3. User Training: Provide online and in-person tutorials for faculty and IT staff.
4. System Ambassadors: Select key representatives in each department to support users during the first month of implementation.
5. Feedback Collection: Use online surveys to gather user opinions and improve the interface.

Key Messages

- Save time and eliminate paperwork.
- Borrow and return equipment more easily.
- Improve accountability and data accuracy.

These efforts will promote a smooth transition from the manual process to a reliable digital platform.

6. ORGANIZATION AND STAFFING

The project will be developed and managed by the IT Department following an Agile development approach, ensuring flexibility and iterative improvements.

Role	Responsibility
Project Sponsor	Approves project budget, monitors progress, and ensures alignment with EIU's strategy. (<i>Head of IT Department</i>)
Project Manager	Oversees daily operations, manages timeline, and coordinates between technical and administrative teams.
Backend Developer	Implements FastAPI services, database logic, and API integration.
Frontend Developer	Designs the user interface, ensuring responsiveness and usability.
QA Tester	Conducts functional, performance, and usability testing.

IT Support Staff	Deploys the system, maintains documentation, and provides user training and technical assistance.
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If necessary, part-time student interns may be involved for front-end or testing support, providing both learning experience and cost reduction.

7. SCHEDULE

The estimated project duration is five months (November 2025 – March 2026).

Phase	Duration (weeks)	Key Deliverables	Milestones
Requirements Analysis	3	Requirements specification	Sign-off on system scope
System Design	4	UI mockups, database schema	Design approval
Development	8	Completed modules & internal test	Core features implemented
User Testing	3	Feedback reports	Pilot testing completed
Deployment & Training	2	System go-live and staff orientation	Official system launch

Estimated Go-Live Date: March 30, 2026

8. FINANCIAL PROJECTIONS

Initial Investment Estimate

Category	Estimated Cost (VND)	Remarks
Software Development	80,000,000	Developer labor and testing tool
Hosting & Cloud Setup	10,000,000	Server or Render/Azure plan
Testing & Training	15,000,000	Faculty and staff training sessions
Documentation & Maintenance	15,000,000	Manuals, updates, and technical support
Total Estimated Cost	120,000,000	N/A

Annual Operational Cost and Benefits

Category	Annual Cost (VND)	Benefit (VND)	Description
System Maintenance & Support	18,000,000	N/A	IT support and updates

Hosting & Email Services	6,000,000	N/A	Cloud operations and communication tools
N/A	N/A	100,000,000	Staff time saved from paperwork
N/A	N/A	40,000,000	Reduced equipment loss and improved monitoring
Total	24,000,000	140,000,000	N/A

Financial Summary

- ROI: $\approx 96.6\%$
- Payback Period: ~ 1.2 years
- Assumptions: Project completed on schedule, internal team development, no external licensing cost.
- Non-financial Benefits: Improved transparency, sustainability, staff satisfaction, and institutional reputation.

9. FINDINGS AND RECOMMENDATIONS

The feasibility study confirms that the EBMS project is technically, operationally, and financially feasible. It aligns with EIU's digital transformation objectives and supports administrative efficiency.

Findings

Technical Feasibility:

The system uses proven, open-source technologies that EIU's IT staff can manage. Technical risks are minimal.

Economic Feasibility:

The initial cost is modest compared to long-term savings and efficiency gains. ROI is positive within the first two years.

Operational Feasibility:

The system fits existing workflows and requires minimal training. Faculty and students are familiar with web applications, ensuring easy adoption.

Schedule Feasibility:

The project timeline is realistic, and most hardware and software resources are already available.

Recommendation

Based on the analysis, the Equipment Borrowing Management System is:

- Technically sound
- Economically beneficial
- Operationally practical

- Schedule-feasible and low risk

Recommendation: *Accept and proceed to full implementation.*

After successful pilot testing, the system should be scaled to manage all equipment and resources within the university.

Implementation Overview

Phase	Duration (weeks)	Key Activities
Initiation	2	Project setup, approvals, budgeting
Analysis & Design	5	Requirement gathering and system design
Development	10	Coding, internal testing, and debugging
Pilot Testing	4	Trial run, feedback collection, issue fixing
Training & Refinement	2	Conduct training sessions and update documentation
Full Deployment	1	Launch across departments

Initial Investment Breakdown

Category	Amount (VND)	Basis of Estimation
Software Development	130,000,000	2 developers × 10M/month × 6.5 months (incl. QA)
Hosting & Cloud Services	8,000,000	Based on EIU Azure annual hosting
Testing & Training	12,000,000	User testing, workshops, and training materials
Licensing & Contingency	15,000,000	Possible licenses and unforeseen expenses
Total Initial Cost	165,000,000	N/A

Annual Operating Costs and Benefits

Category	Amount (VND)	Basis of Estimation
Maintenance	18,000,000	2 IT staff × 5 hrs/month × 12 months × 750,000/hour
Hosting & Email Services	7,000,000	Azure hosting and institutional email
Total Annual Cost	25,000,000	N/A

ROI Calculation

$ROI = (\text{Annual Benefit} - \text{Annual Cost}) / \text{Initial Investment} \times 100$

$ROI = (115M - 25M) / 165M \times 100 \approx 55.8\%$

Payback Period: ~1.7 years

Conclusion:

Given its positive ROI, low technical risk, and strong operational impact, the project should proceed to full implementation and later university-wide deployment after pilot evaluation.

